

Multi-Round Meta-Learning for Label Purification Without Trusted Labels

Mitsuyoshi Otsu, Jun Zhou, Bo Han, Shirui Pan

Abstract—Label noise remains a fundamental challenge in supervised deep learning, particularly as modern neural networks rely on large-scale datasets that frequently suffer from annotation errors due to cost and scalability constraints. While prior approaches address this issue through collaborative training on fully noisy data or meta-learning with access to a small clean subset, the paradigm of clean-label-free label purification remains insufficiently explored, which is particularly critical in expert-annotated domains such as remote sensing and medical imaging, where annotation inconsistencies persist even among trained experts. In this work, we propose a clean-label-free meta-learning framework for label purification. Our approach introduces a multi-round purification strategy that iteratively refines label quality by alternating between base-model training and meta-purification in a decoupled manner, thereby preserving training stability. We further analyze the multi-round purification process under a stochastic contraction perspective and introduce a KL-divergence-based stopping criterion to adaptively select the purification depth. As a result, our framework enables flexible downstream training without reliance on complex noisy-label learning pipelines. Moreover, it captures uncertainty through accumulated label distributions across rounds, improving the reliability of mislabeled instance detection and correction. Extensive experiments on benchmark datasets, including CIFAR and mini-Clothing-1M, show that our method consistently outperforms state-of-the-art techniques. Notably, it achieves up to a 62.61% label accuracy improvement on expert-annotated remote sensing datasets (UC Merced and AID), demonstrating its robustness in real-world, noise-prone environments. The source code is available at <https://github.com/mitsuyoshiohtsu/MRMP>.

Index Terms—Noisy label learning, label purification, meta-learning, generalization bound, early stopping, remote sensing

I. INTRODUCTION

Supervised deep learning has achieved remarkable success across various domains, but its effectiveness hinges on the availability of large volumes of high-quality labeled data. In practice, however, obtaining accurately annotated datasets at scale is often infeasible [1], leading to the adoption of more scalable but error-prone labeling strategies such as crowdsourcing and web scraping [2]. These methods introduce label noise, which can significantly degrade generalization performance. In particular, deep networks are prone to memorization [3], whereby they eventually fit even incorrect labels, causing a substantial drop in test accuracy.

To address this issue, learning with noisy labels (LNL) has emerged as an active research area. Early methods focused on robust loss functions [4], [5] and noise transition matrix

(NTM) estimation [6]–[8], though these approaches often struggle with scalability and accuracy, especially in large-class scenarios. As an alternative, sample selection strategies such as MentorNet [9] and Co-teaching [10] were proposed to prioritize small-loss instances during training. These methods inspired more sophisticated frameworks like DivideMix [11], which combines semi-supervised learning with consistency regularization, and ProMix [12], which improves pseudo-label quality through progressive re-optimization and confidence-weighted label refinement.

Despite their success, most existing methods rely on heuristic small-loss criteria or ensemble-based architectures [10]–[14], which increase computational cost and may exacerbate confirmation bias. Furthermore, data refinement mechanisms—such as instance filtering or pseudo-labeling—are typically embedded as implicit components of the training pipeline [15]–[19], making it difficult to evaluate whether the corrected labels align with human perception or annotation standards.

Another important line of research explores dynamic label correction based on model predictions. One of the earliest methods, Bootstrapping [20], mitigates label noise by linearly combining model predictions with the original labels. Building on this idea, meta-learning approaches such as Meta-Weight-Net [21], Learning to Reweight Examples [22], and Meta Label Correction [23] introduce mechanisms to learn sample weights or correction strategies using a small set of clean validation labels. More recent frameworks, including MetaSLC [24], DMLP [25], and L2B [26], further improve label quality through progressive meta-optimization.

Despite their effectiveness, existing meta-learning methods still rely on clean validation data, which limits their scalability in real-world scenarios. An early attempt to overcome this dependency is PENCIL [27], which integrates label optimization directly into the training objective and achieves moderate improvements without clean supervision. This approach to noisy label purification has gained renewed attention with the emergence of the data-centric AI paradigm [28], which emphasizes label quality and noise mitigation as core principles of robust model development.

These challenges are particularly acute in domains such as medical imaging and remote sensing [29]–[31], where annotations are costly, difficult to validate, and often inconsistent even among experts. However, discussions within the data-centric community have primarily focused on efficient data creation pipelines involving human-in-the-loop processes, rather than on machine learning methodologies capable of achieving downstream performance comparable to model-centric noise-robust learning approaches.

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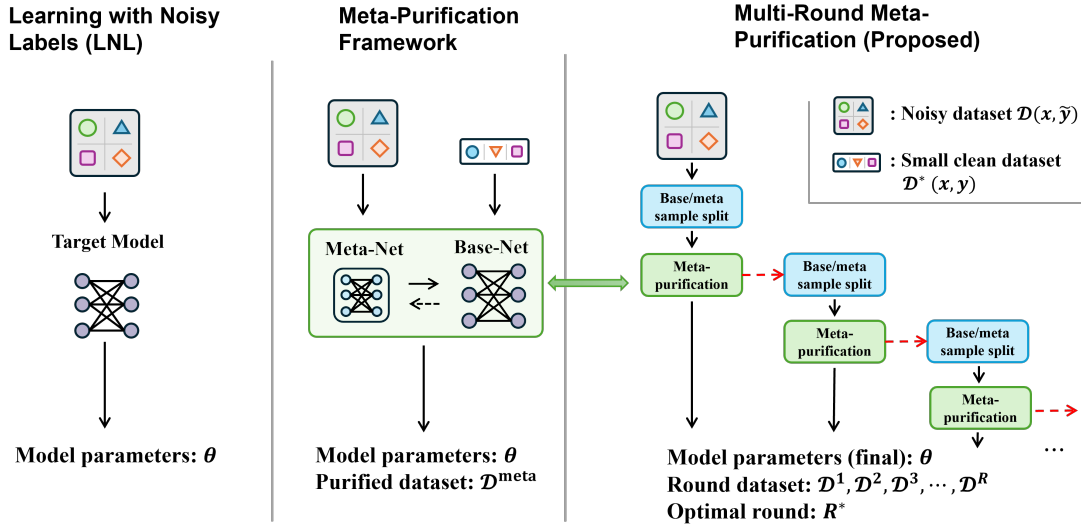


Fig. 1. Comparison of our proposed multi-round, clean-label-free meta-purification framework with prior learning with noisy labels methods and single-round meta-purification approaches that rely on clean data.

In this paper, we propose a clean-label-free label purification framework based on multi-round meta-learning. Motivated by the observation that noisy-label learning often imposes additional constraints on model training, we adopt a decoupled formulation by structuring the learning process into two stages: (1) label cleaning and (2) model optimization. This separation allows label refinement to proceed independently of downstream training, enabling a more flexible and model-agnostic learning paradigm. Our framework focuses on the first stage, aiming to improve label quality prior to training the final model.

In each purification round, we randomly sample a meta-dataset and refine labels by optimizing their alignment with model performance on the held-out meta-set, as illustrated in Fig. 1. The refinement is guided by a model trained on the corresponding base dataset. To improve efficiency, we fine-tune a pre-trained model and perform lightweight meta-optimization, thereby avoiding the overhead of full meta-training. This purification process is repeated iteratively across multiple rounds.

To mitigate overfitting and confirmation bias over rounds, we introduce an early stopping criterion based on purification dynamics—an indicator that captures the stability and progression of label improvements. This criterion is theoretically grounded in a stochastic contraction analysis, which characterizes convergence behavior and provides a principled stopping rule when further refinement may degrade generalization.

We evaluate our framework on CIFAR-10 and CIFAR-100 with synthetic noise, mini-Clothing-1M with real-world noise, and two expert-annotated remote sensing datasets (UC Merced and AID). Experimental results show that our method significantly improves label quality and model performance without requiring clean labels. Compared to prior noisy-label learning and meta-learning-based purification methods, our approach consistently achieves lower correction error and superior robustness across diverse datasets.

The key contributions of this work are summarized as follows:

- We propose a novel clean-label-free label purification framework based on multi-round meta-learning, which progressively refines noisy labels without requiring clean validation data or ensemble-based architectures.
- We leverage high-performance pre-trained models and adopt scalable meta-learning strategies, enabling efficient label refinement with standard-scale neural networks.
- We introduce an early stopping criterion based on purification dynamics and provide theoretical justification via a stochastic contraction analysis, which guides the selection of the optimal number of refinement rounds.
- We conduct extensive experiments on both synthetic and real-world noisy datasets, demonstrating that our method consistently outperforms or matches state-of-the-art baselines, particularly on expert-annotated remote sensing benchmarks.
- In addition, further experiments show that our framework produces uncertainty-aware label distributions, which can assist human rechecking and support additional refinement when necessary.

II. RELATED WORK

Training deep neural networks with noisy labels remains a significant challenge, as deep models tend to memorize incorrect labels during the later stages of training [3]. Various strategies have been developed to address this issue, including robust training procedures and data purification techniques.

A. Learning with Noisy Labels

As a pioneering approach, MentorNet [9] introduced curriculum-based sample selection by training a mentor network to weight training samples according to their difficulty. Building on this idea, Co-teaching [10] proposed training two

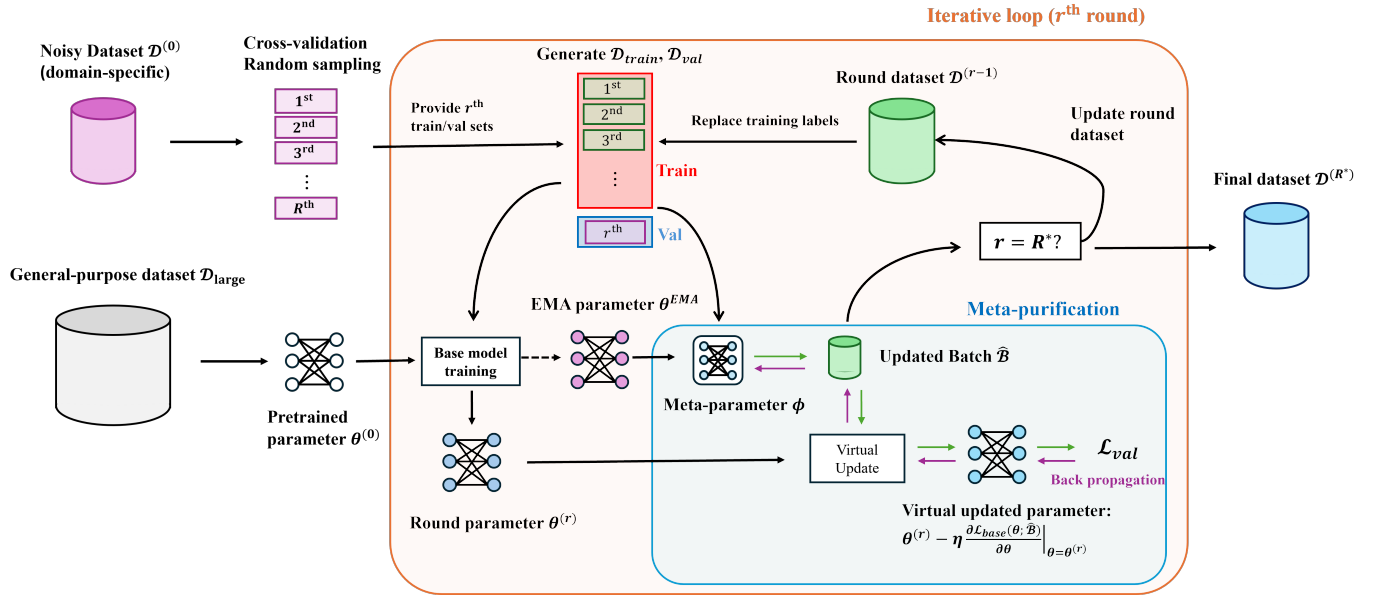


Fig. 2. Overview of the proposed Multi-Round Meta-Purification (MRMP) framework. In each round r , meta-parameter ϕ is optimized to generate refined labels via meta-purification, using validation loss $\mathcal{L}_{\text{val}}$ through virtual updates of the trained base model. Refined labels are propagated iteratively until the early stopping round R^* .

networks simultaneously, where each network selects small-loss samples for the other, thereby alleviating confirmation bias. Co-teaching+ [32] further improves sample diversity by explicitly encouraging prediction disagreement between the two networks, reducing the risk of confirmation bias caused by overly similar representations. In contrast, JoCoR [13] enhances robustness by incorporating prediction consistency into the loss function, enabling joint learning that is more resilient to label noise.

As a combination of co-training and semi-supervised learning, DivideMix [11] advances noisy-label learning by modeling per-sample losses using a Gaussian Mixture Model (GMM). It treats low-loss samples as labeled and high-loss samples as unlabeled, and trains two networks collaboratively using MixUp [33]. While DivideMix achieves strong performance, it introduces considerable training complexity and relies heavily on accurate loss modeling. ProMix [12] extends DivideMix by progressively reweighting pseudo-labels based on prediction consistency across epochs, thereby improving label reliability without explicitly partitioning samples into labeled and unlabeled sets. This progressive refinement strategy reduces sensitivity to early-stage prediction errors.

From a different perspective, pseudo-labeling strategies aim to refine noisy annotations throughout training. Bootstrapping [20] stabilizes supervision by interpolating between ground-truth labels and model predictions, while PENCIL [27] treats labels as optimization variables that are updated jointly with model parameters. ELR [18] mitigates overfitting to noisy labels by enforcing consistency between current predictions and an exponential moving average of historical outputs. However, confidence-based update mechanisms remain vulnerable under high-noise regimes, where prediction certainty is inherently unreliable.

B. Meta-Learning-Based Label Purification

Meta-learning approaches adapt sample importance or label correction functions dynamically to improve robustness against label noise. L2RW [22] learns per-sample weights by minimizing the meta-validation loss on a small clean dataset, while Meta-Weight-Net [21] parameterizes a weighting function based on training losses and optimizes it via meta-gradients. Meta Label Correction (MLC) [34] extends these ideas by learning a separate label correction network through bi-level optimization. Although effective, these methods rely on clean validation sets and incur significant computational overhead due to nested gradient computations. To reduce clean data requirements, MLNT [35] simulates noisy meta-learning episodes to enhance generalization. However, the joint optimization of representations and labels often destabilizes training, particularly in high-noise scenarios.

Recent meta-learning-based methods aim to address these challenges more effectively. The Decoupled Meta Label Purifier (DMLP) [25] separates feature learning and label correction into two distinct stages. DMLP first trains a label-agnostic feature extractor and then refines noisy labels using a lightweight meta-label corrector. It introduces Intrinsic Primary Correction (IPC) for global refinement based on instance similarities and Extrinsic Auxiliary Correction (EAC) for fine-grained correction of residual errors. This decoupled structure stabilizes optimization, reduces computational cost, and improves purification effectiveness without relying on clean validation sets.

L2B [26] further advances meta-learning-based purification by jointly reweighting sample labels and instances through a learned bootstrapping mechanism guided by meta-objectives. It adapts supervision dynamically during training and leverages a small clean validation set to optimize the reweighting functions via meta-gradient updates.

Compared to sample selection, pseudo-labeling, and co-training methods that often filter out uncertain samples, meta-learning-based label purification approaches—such as DMLP and L2B—aim to refine all available labels. This strategy maximizes data utilization and significantly enhances robustness under varying noise conditions. However, most existing meta-learning methods still require clean validation data and involve complex gradient-based optimization, which limits their scalability in practical applications.

III. METHODOLOGY

This section introduces our multi-round meta-purification framework for learning with noisy labels and presents the theoretical foundation for optimizing the number of purification rounds. The proposed approach iteratively refines noisy labels by leveraging pseudo-labels predicted by a fine-tuned model trained on previously refined labels. We first describe the overall structure of the framework, followed by details of the noisy meta-learning mechanism used for label correction. Finally, we present a stochastic contraction analysis that provides theoretical support for the multi-round process, including a formulation for determining the optimal number of refinement rounds.

A. Overall Framework: Multi-Round Meta-Purification

We propose a multi-round meta-purification (MRMP) framework designed to improve robustness against label noise. The core idea is to iteratively refine labels through a decoupled process that separates model training and label correction. This decoupling offers two main advantages: (1) it prevents the leakage of noisy information from meta-validation data into the base model, and (2) it maintains high model performance by avoiding the instability commonly introduced by bi-level optimization in meta-learning. To further enhance performance, the framework leverages a high-performance pre-trained model for initialization, providing strong representational capacity. The model's predictions are then passed to a meta-network that performs label purification through a memory-efficient meta-learning strategy.

Each purification round in the MRMP framework comprises two key stages: (i) fine-tuning the base model to stabilize predictions using an Exponential Moving Average (EMA), and (ii) optimizing a meta-network based on EMA-stabilized predictions via virtual updates of the fixed base model.

In stage (i), the base model is trained on the noisy training set for the current round, initialized from a high-performance pre-trained model to ensure strong representational capacity. Concurrently, an EMA model is updated in parallel to generate temporally stable predictions, which serve as reference signals during the subsequent meta-optimization stage.

In stage (ii), the meta-network g_ϕ is optimized using meta-gradients computed from the validation loss. During this process, the base model remains fixed to prevent optimization instability caused by noisy pseudo-labels. After meta-optimization, the refined labels produced by g_ϕ are used to update the training set, and the framework proceeds to the next round with re-initialized model parameters. This two-stage

Algorithm 1 Multi-Round Meta-Purification (MRMP)

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1: Input: Noisy dataset  $\mathcal{D}^{(0)} = \{(x_i, \tilde{y}_i)\}_{i=1}^n$ , pretrained
   weights  $\theta^{(0)}$ , number of rounds  $R$ 
2: Output: Final base model  $f^*$  and purified dataset  $\mathcal{D}^{(R)}$ 
3: for round  $r = 1$  to  $R$  do
4:   Sample  $\mathcal{D}_{\text{train}} \subset \mathcal{D}^{(r-1)}$ , disjoint  $\mathcal{D}_{\text{val}} \subset \mathcal{D}^{(0)}$ 
5:   Reset  $f_\theta \leftarrow \theta^{(0)}$ , initialize meta-network  $g_\phi$ 
6:   Train  $f_\theta$  on  $\mathcal{D}_{\text{train}}$  and update EMA teacher  $\hat{f}^{(r)}$ 
7:   for all  $(x_i, \tilde{y}_i) \in \mathcal{D}_{\text{train}}$  do
8:     Predict pseudo-label:  $\hat{y}_i \leftarrow g_\phi(\hat{f}^{(r)}(x_i), \tilde{y}_i)$ 
9:     Virtual update:  $f'_\theta \leftarrow f_\theta - \alpha \nabla_{\theta} \mathcal{L}(f_\theta(x_i), \hat{y}_i)$ 
10:    Compute validation loss on  $\mathcal{D}_{\text{val}}$  using  $f'_\theta$ 
11:    Update  $g_\phi$  via meta-gradient descent
12:   end for
13:   for all  $(x_i, \tilde{y}_i) \in \mathcal{D}_{\text{train}}$  do
14:     Refine label:  $\tilde{y}_i \leftarrow g_\phi(\hat{f}^{(r)}(x_i), \tilde{y}_i)$ 
15:   end for
16:   Set  $\mathcal{D}^{(r)} \leftarrow \mathcal{D}^{(r-1)}$  with updated labels for  $\mathcal{D}_{\text{train}}$ 
17: end for
18: Reset  $f_\theta \leftarrow \theta^{(0)}$ 
19: Train  $f_\theta$  on  $\mathcal{D}^{(R)}$  to obtain  $f^*$ 
20: Return:  $f^*$  and  $\mathcal{D}^{(R)}$ 

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process is repeated iteratively until label quality converges or a predefined number of purification rounds R is completed.

The complete procedure is summarized in Algorithm 1 and illustrated in Fig. 2. At each round, the noisy dataset is randomly partitioned into a base training set and a meta-validation set to facilitate meta-learning, promote task diversity, and prevent overfitting to a fixed data split. To avoid information leakage across rounds, all model parameters are re-initialized at the start of each round. Moreover, the validation set $\mathcal{D}_{\text{val}}$ is consistently sampled from the original dataset $\mathcal{D}^{(0)}$, ensuring that the purification process is driven solely by the evolving base dataset, independent of accumulated model memory or validation drift. This design ensures a clean separation between rounds and maintains the integrity of the label refinement process.

B. Meta-Purification Process Within Each Round

Each purification round in the proposed MRMP framework incorporates a meta-learning step to refine noisy training labels using a disjoint noisy validation set. Unlike conventional meta-learning methods that assume access to clean validation data, our approach acknowledges that both the training and validation sets may contain label noise. This setting necessitates a clean-label-free meta-purification strategy capable of mitigating confirmation bias throughout the iterative meta-optimization process.

The label refinement process is formulated as a bi-level optimization problem:

$$\min_{\phi} \mathbb{E}_{(x, \tilde{y}) \sim \mathcal{D}_{\text{val}}} [\ell(f_{\theta^*(\phi)}(x), \tilde{y})];$$

$$\theta^*(\phi) = \arg \min_{\theta} \mathbb{E}_{(x, \tilde{y}) \sim \mathcal{D}_{\text{train}}} [\ell(f_{\theta}(x), g_{\phi}(x, \tilde{y}))]$$

Here, $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{val}}$ denote the noisy training and validation sets, respectively. The meta-network g_ϕ refines the noisy label $\tilde{y}$ to produce improved supervision for the base model f_θ .

Let $\theta^{(t)}$ represent model parameters at training epoch t , and define the learned feature representation as $h^{(t)}(x) = f_{\theta^{(t)}}(x)$. For simplicity, we focus on two temporal states: an early-stage model $\theta^{(1)}$, which captures generalizable patterns, and a later-stage overfitted model $\theta^{(2)}$. The representational difference,

$$\Delta h(x) = h^{(2)}(x) - h^{(1)}(x),$$

captures the shift in features due to memorization of noisy labels [3], [18]. Because $\theta^{(1)}$ is more likely to preserve clean-signal structure, the directional change $\Delta h(x)$ serves as a proxy for label corruption—indicating transitions from clean to memorized noisy labels.

To leverage this insight, we define the meta-network to generate pseudo-labels by exploiting the representational dynamics between two training stages:

$$g_\phi(h^{(1)}(x), h^{(2)}(x), \tilde{y}) \approx y^*,$$

where y^* denotes the latent clean label. By capturing the discrepancy between early and later feature representations, the meta-network g_ϕ can estimate the direction of label corruption and propose corrected supervision accordingly.

Since directly retaining the reliable early-stage model $\theta^{(1)}$ is often impractical without access to clean labels, we instead approximate it using an Exponential Moving Average (EMA) of model parameters:

$$\hat{\theta}_t^{\text{EMA}} = \beta \hat{\theta}_{t-1}^{\text{EMA}} + (1 - \beta) \theta^{(t)},$$

where $\beta \in (0, 1)$ is a momentum coefficient. The EMA model $f_{\hat{\theta}_t^{\text{EMA}}}(x)$ smooths temporal fluctuations in model updates, yielding a more stable and generalizable feature extractor for downstream meta-optimization.

To enhance generalization, we initialize the base model with pre-trained weights $\theta^{(0)}$ obtained from a large-scale dataset such as ImageNet. From this starting point, the model undergoes the representational transition:

$$f_{\theta^{(0)}}(x) \rightarrow f_{\theta^{(t)}}(x),$$

which reflects a gradual shift from general-purpose features to task-specific embeddings that may increasingly encode noise-induced patterns. Guided by this condition, the EMA model acts as a temporally smoothed estimator of the training trajectory, offering reliable signals for detecting noisy labels. Empirically, we observe that the later-stage features $f_{\theta^{(t)}}(x)$ tend to align closely with the noisy labels $\tilde{y}$ in the final training stages. Thus, we approximate $f_{\theta^{(t)}}(x) \approx \tilde{y}$ and simplify the meta-network input to $g_\phi(f_{\hat{\theta}_t^{\text{EMA}}}(x), \tilde{y})$, thereby preserving temporal guidance while reducing computational complexity.

Regarding the meta-network, we incorporate a cross-attention mechanism into g_ϕ to assess the consistency between the noisy label $\tilde{y}$ and the prediction from the EMA model. The attention inputs are defined as:

$$q = W_q \tilde{y}, \quad k = W_k f_{\hat{\theta}_t^{\text{EMA}}}(x), \quad v = f_{\hat{\theta}_t^{\text{EMA}}}(x), \quad (1)$$

where $W_q, W_k \in \mathbb{R}^{d \times c}$ are trainable projection matrices, and d and c represent the dimensions of the attention and feature

spaces, respectively. Unlike standard attention modules, we omit trainable value projections and directly use the raw EMA features as values to mitigate training instability caused by the sensitivity of noisy labels to model updates. This design enables the meta-network to attend to reliable signals and produce context-aware corrections without overfitting to label noise. The attention response is computed as:

$$\alpha(q, k, v) = \sigma \left(-\frac{q^\top k}{\tau} \right) v, \quad (2)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function and $\tau > 0$ is a temperature parameter controlling the sharpness of the response. To improve numerical stability, we omit the standard $\sqrt{d}$ scaling by choosing a sufficiently small attention dimension d . The use of a negative sign emphasizes inconsistency: when the prediction and the noisy label disagree, the attention score increases, thereby promoting correction. In contrast, stronger agreement results in lower attention weights.

The refined label is subsequently generated via residual correction:

$$g_\phi(x, \tilde{y}, f_{\hat{\theta}_t^{\text{EMA}}}(x)) = \text{Norm}(\tilde{y} + \alpha(q, k, v)), \quad (3)$$

where $\text{Norm}(\cdot)$ denotes simplex normalization, ensuring that the refined output remains a valid probability distribution. This formulation preserves the original label signal while allowing data-dependent adjustments informed by semantic alignment.

Furthermore, to prevent meta-training from being adversely affected by erroneous supervision, we adopt the penalty loss proposed in DivideMix [11] during base-model training. This loss explicitly discourages over-confident predictions on ambiguous samples, thereby preserving predictive uncertainty that is essential for reliable meta-level label refinement.

C. Theoretical Analysis of the Optimal Round in Multi-Round Purification

We analyze the evolution of multi-round label purification and derive a principled criterion for selecting the optimal stopping round without access to clean labels. Let $\tilde{y}^{(t)}$ denote the refined soft labels at purification round t , and define the residual noise level

$$\delta^{(t)} := \mathbb{E} \|\tilde{y}^{(t)} - y^*\|_1,$$

where y^* denotes the unknown clean label distribution.

Consistent with generalization bounds in multi-task meta-learning [36], repeated pseudo-task resampling encourages the decorrelation of instance-dependent noise patterns across rounds, while meta-learning stability mitigates their potential amplification. The formal justification and detailed assumptions underlying this behavior are provided in the supplementary material (Section S.1). Our goal is to characterize the evolution of $\delta^{(t)}$ and determine the optimal purification depth

$$R^* := \arg \min_t \delta^{(t)}.$$

Assumption 1. The refinement process satisfies a stochastic contraction with additive error:

$$\delta^{(t+1)} \leq \Gamma^{(t)} \delta^{(t)} + \bar{\epsilon}^{(t)}, \quad \mathbb{E}[\Gamma^{(t)}] = \gamma < 1, \quad (4)$$

where $\Gamma^{(t)} \in [0, 1]$ is a random contraction factor and $\bar{\epsilon}^{(t)} \geq 0$ captures irreducible approximation and optimization errors.

Assumption 1 induces a contraction–error trade-off: the factor $\gamma < 1$ drives geometric decay, while the accumulated error $\bar{\epsilon}^{(t)}$ imposes a lower bound. Thus, $\delta^{(t)}$ decreases initially and stabilizes once the error floor dominates, implying a well-defined optimal depth R^* . Since $\delta^{(t)}$ depends on the unknown clean labels, it is not directly observable. To obtain a computable surrogate, we introduce the inter-round KL divergence

$$f(t) = \frac{1}{n} \sum_{i=1}^n \text{KL}(\tilde{y}_i^{(t+1)} \| \tilde{y}_i^{(t)}),$$

which measures the magnitude of refinement updates between consecutive rounds and can be evaluated during training. The following result shows that the KL divergence controls the variation of the unobservable residual noise.

Theorem 1. For all refinement rounds t ,

$$|\delta^{(t+1)} - \delta^{(t)}| \leq \sqrt{2f(t)}.$$

Proof. By definition,

$$\delta^{(t)} = \mathbb{E} \|\tilde{y}^{(t)} - y^*\|_1.$$

Applying the reverse triangle inequality gives

$$|\|\tilde{y}^{(t+1)} - y^*\|_1 - \|\tilde{y}^{(t)} - y^*\|_1| \leq \|\tilde{y}^{(t+1)} - \tilde{y}^{(t)}\|_1.$$

Taking expectations yields

$$|\delta^{(t+1)} - \delta^{(t)}| \leq \mathbb{E} \|\tilde{y}^{(t+1)} - \tilde{y}^{(t)}\|_1.$$

Applying Pinsker's inequality sample-wise and Jensen's inequality establishes the bound. $\square$

Theorem 2. Under Assumption 1, selecting the earliest round at which $f(t)$ reaches a stable minimal plateau yields a consistent estimate of the optimal purification depth:

$$\hat{R}^* \rightarrow R^*.$$

Proof. By Theorem 1, the variation of $\delta^{(t)}$ is bounded by $\sqrt{2f(t)}$. When $f(t)$ stabilizes near zero, further changes in $\delta^{(t)}$ are negligible. Since $\delta^{(t)}$ initially decreases under contraction and later stabilizes due to accumulated error, the KL plateau coincides with the optimal stopping regime. $\square$

Together, Theorems 1 and 2 show that the observable KL trajectory governs the evolution of the unobservable residual noise and provides a principled stopping rule. In particular, small $f(t)$ implies negligible variation in $\delta^{(t)}$, so the KL curve reflects the contraction–error transition underlying Assumption 1 and marks the onset of saturation. To derive a practical estimator of R^* , we characterize the decay–error structure of $\delta^{(t)}$. Under Assumption 1,

$$\delta^{(t)} = \delta^{(0)}\gamma^t + \sum_{s=0}^{t-1} \bar{\epsilon}^{(s)},$$

where the first term decays geometrically and the second accumulates sublinearly (see Supplementary Section S.1).

Assumption 2. The KL sequence $\{f(t)\}_{t \geq 0}$ evolves smoothly in the sense that

$$|f(t+1) - f(t)| \leq L_f + \xi_t, \quad \mathbb{E}[\xi_t] = 0, \quad |\xi_t| \leq \sigma,$$

where ξ_t represents bounded stochastic fluctuations.

Theorem 3. Under Assumptions 1–2, the KL trajectory admits a structured decomposition

$$f(t) = f_{\text{decay}}(t) + f_{\text{floor}}(t),$$

where $f_{\text{decay}}(t)$ is monotone decreasing and proportional to $\delta^{(0)}\gamma^t$, while $f_{\text{floor}}(t)$ is slowly varying and proportional to the accumulated approximation error.

Proof. By Theorem 1, $f(t)$ scales with $|\delta^{(t+1)} - \delta^{(t)}|^2$. Substituting the decomposition of $\delta^{(t)}$ obtained from (4) yields two distinct contributions: a geometrically decaying term induced by the contraction factor γ , and a residual term driven by the accumulated approximation errors. Assumption 2 guarantees that stochastic fluctuations do not introduce independent oscillations, so the KL trajectory inherits the same decay–saturation structure. $\square$

Theorem 3 explains the empirical shape of the KL curve: early rounds are dominated by contraction-driven improvement, producing rapid decay, whereas later rounds are governed by residual error, resulting in a stable floor. This decay–floor structure underlies the emergence of a plateau and enables principled estimation of the optimal depth.

Assuming an exponential decay and a constant error floor,

$$f_{\text{decay}}(t) \approx \alpha e^{-\lambda t}, \quad f_{\text{floor}}(t) \approx \eta,$$

the optimal purification depth satisfies

$$f_{\text{decay}}(R^*) \approx f_{\text{floor}}(R^*),$$

leading to the estimator

$$\hat{R}^* = \left\lfloor \frac{1}{\lambda} \log\left(\frac{\alpha}{\eta}\right) \right\rfloor + 1.$$

This estimator operationalizes the same contraction–error balance that defines R^* and exploits the global shape of the KL trajectory rather than local fluctuations, making it robust to transient stochastic noise. Its validity relies on the following conditions: (i) the refinement dynamics exhibit stochastic contraction in expectation, (ii) the KL trajectory evolves smoothly up to bounded fluctuations, (iii) the expected contraction factor satisfies $\gamma < 1$, and (iv) approximation errors accumulate sublinearly so that a saturation plateau emerges. Even when these conditions hold only approximately—as is typical in practical training—the KL trajectory empirically preserves the predicted contraction–saturation pattern, enabling $\hat{R}^*$ to reliably identify a near-optimal stopping round.

IV. EXPERIMENTS

A. Datasets and Settings

To evaluate the effectiveness of the proposed Multi-Round Meta-Purification (MRMP) framework, we conduct experiments on both synthetic and real-world noisy datasets. The selected benchmarks include both standard computer vision tasks and expert-annotated remote sensing datasets, enabling a comprehensive evaluation under diverse label noise conditions. We consider the following datasets:

CIFAR & CIFAR-N: These datasets contain 60,000 images of size 32×32 from 10 and 100 categories. Following standard

TABLE I

PURIFIED LABEL ACCURACY (%) ON CIFAR-10/100 WITH SYNTHETIC AND CIFAR-10N/100N WITH HUMAN-ANNOTATED NOISE. CIFAR-10 USES SYMMETRIC (20%, 50%, 70%) AND ASYMMETRIC (40%) NOISE, WHILE CIFAR-100 IS EVALUATED UNDER SYMMETRIC NOISE ONLY. “NEEDS CLEAN VAL?” INDICATES WHETHER A SMALL CLEAN VALIDATION SET IS REQUIRED.

Method	Needs clean val?	CIFAR-10				CIFAR-100			CIFAR-10N	CIFAR-100N
		20%	50%	70%	40% (Asym.)	20%	50%	70%	Worst	Noisy Fine
No process	–	82.03	55.07	37.00	80.25	80.12	50.36	30.99	59.79	59.80
Bootstrapping [20]	No	91.77	83.10	65.48	81.43	83.67	61.04	31.13	77.33	64.38
PENCIL [27]	No	92.97	80.08	60.87	81.67	86.92	59.49	28.36	78.83	63.28
ELR [18]	No	94.11	85.99	76.79	86.89	86.19	68.21	46.66	84.00	68.28
DivideMix [11]	No	88.24	82.69	80.95	83.89	76.72	69.75	58.65	82.34	66.34
ProMix [12]	No	95.27	94.02	85.86	92.62	87.55	80.66	65.20	91.58	68.13
GFWS [38]	No	96.68	94.55	92.03	93.66	84.84	76.44	64.91	92.13	68.63
WSC [39]	No	97.88	95.85	93.27	90.38	88.76	80.14	69.29	92.92	73.16
L2B [26]	Yes	83.12	59.24	42.74	82.38	74.61	48.16	28.99	59.69	50.46
DMLP [25]	Yes	92.29	92.15	91.83	92.12	84.00	66.10	57.50	91.32	60.88
MRMP (Ours)	No	98.85	98.13	94.24	92.80	95.65	87.85	89.51	92.53	79.48

TABLE II

PURIFIED LABEL ACCURACY (%) ON MINI-CLOTHING-1M WITH REAL-WORLD HUMAN-ANNOTATED NOISE.

Method	No process	Bootstrapping [20]	PENCIL [27]	ELR [18]	DivideMix [11]	ProMix [12]	WSC [39]	MRMP
Label Acc. (%)	61.73	61.73	61.73	62.44	65.93	63.16	66.28	67.46

practice [11], we construct synthetic noisy-label settings, with symmetric noise applied at rates of 20%, 50%, and 70% via uniform label flipping, and asymmetric noise generated using the class-dependent transition protocol in [11] at a noise rate of 40% on CIFAR-10. We also evaluate on CIFAR-10N/100N [37], which provides human-annotated noisy labels for studying instance-dependent noise.

mini-Clothing-1M: Derived from Clothing-1M, which includes over one million real-world noisy samples, we construct a 37,497-sample subset containing both noisy and clean labels—termed *mini-Clothing-1M* (details in Supplementary Material S.2). This subset is substantially more challenging due to its significantly reduced scale, compared to the original dataset.

UC Merced & AID: UC Merced and AID contain 2,100 and 10,000 aerial images across 21 and 30 classes. We introduce symmetric and asymmetric synthetic noise following the CIFAR protocol; asymmetric flips occur among semantically or visually similar classes (details in Supplementary Material S.2). This provides reproducible benchmarks for remote-sensing-specific noise.

Experimental Setting: We adopt the ConvNeXt architecture family [40] with ImageNet-pretrained weights to ensure sufficient model capacity under varying noise conditions. ConvNeXt-Small is used for all CIFAR-10/100 experiments, except for the most challenging setting (CIFAR-100 with 70% symmetric noise), where ConvNeXt-Base is employed. ConvNeXt-Base is also adopted for CIFAR-10N/100N, mini-Clothing-1M, and the UC Merced and AID datasets. In addition, we evaluate RemoteCLIP with a ViT-B-32 backbone as a domain-specific backbone for remote sensing [41].

All images are resized to 224×224 . Standard data augmentations are applied, including random cropping and horizontal flipping for CIFAR and mini-Clothing-1M, and horizontal and vertical flipping for UC Merced and AID. The default hyperparameters are set to $\beta = 0.90$ and $\tau = 0.7$, with

sensitivity analysis reported in Supplementary Material S.6. To ensure memory efficiency, we adopt the central difference approximation following the SAMA strategy [42], which avoids computing second-order gradients during meta-learning. (Further implementation details are provided in Supplementary Material S.3.) All experiments are conducted using a single NVIDIA RTX A5000 GPU.

Evaluation metrics: In addition to the standard classification accuracy of the trained models, we evaluate label purification quality using the *purified label accuracy*,

$$\text{Acc}_{\text{label}} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\tilde{y}_i = y_i^*),$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, $\tilde{y}_i$ is the refined (purified) label for the i -th sample, and y_i^* is its corresponding clean label. This metric measures the proportion of correctly purified labels and directly reflects the effectiveness of the label refinement process.

B. Experimental Results

We evaluate label purification performance on five benchmark datasets with both synthetic and real-world label noise: CIFAR-10, CIFAR-100, CIFAR-10N/100N, mini-Clothing-1M, UC Merced, and AID. For fair comparison, we benchmark our method against several state-of-the-art approaches designed for learning with noisy labels.

1) CIFAR & CIFAR-N: We report purified label accuracy under synthetic noise on CIFAR-10/100, as well as under human-annotated instance-dependent noise on CIFAR-10N/100N. The comparison includes label refinement methods without clean supervision (Bootstrapping [20], PENCIL [27], ELR [18]), semi-supervised noise-robust methods (DivideMix [11], ProMix [12]), state-of-the-art noise-robust approaches (GFWS, WSC), and meta-purification frameworks requiring clean validation data (L2B [26] and DMLP [25]).

TABLE III
PURIFIED LABEL ACCURACY (%) ON UC MERCED AND AID UNDER SYMMETRIC (20%, 50%, 70%) AND ASYMMETRIC (40%) LABEL NOISE.

Method	UC Merced				AID			
	20%	50%	70%	40% (asym)	20%	50%	70%	40% (asym)
no process	81.010	53.510	33.000	80.650	80.550	50.990	32.000	86.720
Bootstrapping [20]	85.893	75.298	59.643	86.131	85.825	76.763	59.013	89.500
PENCIL [27]	90.595	70.417	51.190	81.845	84.888	52.450	32.513	86.738
ELR [18]	89.167	64.048	40.833	85.119	89.988	63.950	35.638	93.075
DivideMix [11]	96.786	93.393	67.083	91.369	93.500	89.800	76.483	88.238
ProMix [12]	87.619	88.393	64.524	81.607	60.119	59.583	57.143	71.429
WSC [39]	86.964	59.048	38.512	77.619	95.575	82.175	53.662	91.237
MRMP w/ ConvNeXt-B	98.393	96.131	89.107	88.155	97.850	95.263	92.225	93.450
MRMP w/ RemoteCLIP	98.988	95.893	93.512	96.548	98.038	94.650	94.613	98.350

MRMP achieves the highest label accuracy in the majority of settings. Under symmetric noise on CIFAR-10, it maintains top performance, outperforming the second-best method (WSC) by approximately **1–2%**, and achieving up to **+57.2%** relative improvement over the no-processing baseline. On CIFAR-100, MRMP shows strong robustness, outperforming the second-best method by about **+7%** under 20% and 50% symmetric noise, and by more than **+20%** at the 70% noise level. Under instance-dependent human noise on CIFAR-10N/100N, MRMP remains competitive with leading methods. In particular, it achieves **+6%** higher label accuracy than the second-best method on CIFAR-100N. While its performance on CIFAR-10N is comparable to WSC, which attains the highest accuracy, MRMP exhibits advantages in mitigating confirmation bias, a property that is critical for practical deployment (see Section V.A).

2) *mini-Clothing-1M*: To assess performance under real-world label noise, we evaluate MRMP on the mini-Clothing-1M dataset, a challenging pruned subset of Clothing-1M in which each instance is paired with both noisy and clean labels. As summarized in Table II, MRMP achieves the highest purified label accuracy of **67.46%**, representing a **+5.73%** improvement over the no-processing baseline and outperforming all comparison methods. Among existing approaches, WSC attains the strongest competing performance (66.28%), followed by DivideMix and ProMix, while ELR yields more moderate improvement. In contrast, earlier correction-based methods such as Bootstrapping and PENCIL exhibit limited gains and remain close to the no-processing baseline.

These results indicate that effective label purification on mini-Clothing-1M requires explicit control of biased supervision signals to prevent the accumulation of misleading information during training. Given the limited number of samples, strong class imbalance, and severe annotation noise, overconfident updates and bias amplification can be induced in conventional learning pipelines. In contrast, MRMP employs multi-round refinement with implicit smoothing, enabling low-bias label updates. This behavior is critical in noise-dominated, data-scarce settings where aggressive correction can distort, rather than purify, the data distribution.

3) *UC Merced and AID*: Table III reports purified label accuracy on two expert-annotated remote sensing datasets—UC Merced and AID—under symmetric (20%, 50%, 70%) and asymmetric (40%) label noise. MRMP consistently achieves

the highest performance across all settings, with particularly strong gains under severe noise. For example, on UC Merced with 70% symmetric noise, MRMP with a ConvNeXt-B backbone achieves 89.11%, exceeding DivideMix (67.08%) by over **+22%**, while replacing the backbone with the domain-specific foundation model RemoteCLIP [41] further boosts performance to **93.51%**. On AID with the same noise level, MRMP attains 92.23% with ConvNeXt-B and improves to **94.61%** with RemoteCLIP, outperforming DivideMix (76.48%) by approximately **+18%**.

While DivideMix remains competitive under mild to moderate noise, ProMix exhibits substantial degradation on both datasets, despite its strong performance on natural image benchmarks such as CIFAR-10/100. This suggests that methods relying on extensive augmentations and multiple forward passes may suffer from optimization instability when applied to high-resolution or domain-specific remote sensing imagery.

In contrast, MRMP adopts a decoupled meta-learning design that separates base-model training from label refinement, enabling stable purification across different backbones. Notably, under asymmetric noise settings, incorporating RemoteCLIP consistently improves performance, achieving **96.55%** on UC Merced and **98.35%** on AID at the 40% noise level, corresponding to improvements of around **+5%** relative to the second-best method. Overall, MRMP delivers up to **+62.61%** improvement over the unprocessed baseline, demonstrating its effectiveness and scalability for expert-annotated, noise-prone remote sensing datasets.

4) *Final Training*: Table IV reports classification accuracy (%) on CIFAR-10 and CIFAR-100 under symmetric (20%, 50%) and asymmetric (40%) label noise. We adopt a standardized training setup using Wide-ResNet-28-10 with RandAugment [49], a widely used augmentation strategy for clean CIFAR training. To ensure fair comparison, we exclude methods that rely on multi-network ensembling.

MRMP consistently achieves top performance across all noise settings. On CIFAR-10, it obtains 97.0% and 97.2% accuracy under 20% and 50% symmetric noise, respectively, outperforming state-of-the-art baselines such as DULC, OT-Filter, and DMLP, whose accuracies are in the range of 94–96% and 94–95%, respectively. Under 40% asymmetric noise, although GFWS and DULC achieve the highest accuracy (95.2%), MRMP attains a competitive accuracy of 93.1%, comparable to DMLP, CMW-Net, and ELR+, whose perfor-

TABLE IV
CLASSIFICATION ACCURACY (%) ON MODELS TRAINED USING PURIFIED LABELS FOR CIFAR-10 AND CIFAR-100 UNDER SYMMETRIC (20%, 50%) AND ASYMMETRIC (40%) LABEL NOISE. RESULTS FOR BASELINE METHODS ARE TAKEN FROM PRIOR WORKS [18], [25], [26], [43]–[45].

Method	CIFAR-10			CIFAR-100	
	20%	50%	40% (asym)	20%	50%
CE	86.9	81.3	85.0	59.6	44.4
Mixup [33]	95.6	85.7	–	67.8	57.3
Bootstrapping [20]	85.2	79.2	–	61.8	45.8
PENCIL [27]	92.4	89.1	91.2	69.4	57.5
REED [46]	95.8	95.6	92.3	76.7	73.0
ELR+ [18]	94.6	93.8	92.7	77.5	72.4
CMW-Net [45]	91.1	–	92.7	70.1	–
BadLabel [44]	95.4	–	91.8	77.4	–
TCL [47]	95.0	93.9	92.6	78.0	73.3
GFWS [38]	95.8	94.6	95.2	79.2	77.5
WSC [39]	96.8	96.6	94.8	77.5	77.5
OT-Filter [43]	96.0	95.3	95.1	76.7	73.8
DULC [48]	96.6	96.0	95.2	79.4	76.4
MW-Net [21]	89.8	–	88.7	66.7	–
L2RW [22]	90.6	85.3	89.2	67.8	55.6
L2B [26]	92.2	88.5	91.8	71.8	64.5
MLC [23]	92.5	88.1	–	66.8	52.7
MSLC [24]	93.4	89.9	91.6	72.5	65.4
DMLP [25]	94.7	94.2	93.9	72.7	68.0
MRMP (Ours)	97.0	97.2	93.1	82.2	79.6

mance falls in the 92–94% range.

On the more challenging CIFAR-100 benchmark, MRMP outperforms all comparison methods by a clear margin, reaching 82.2% and 79.6% accuracy under 20% and 50% symmetric noise. These results represent improvements of **approximately +3% and approximately +2%** over the strongest competing methods, DULC (79.4%) and GFWS (77.5%), at the corresponding noise levels.

Notably, MRMP surpasses prior meta-learning-based approaches such as MW-Net, MSLC, and DMLP despite not relying on clean validation data. This indicates that meta-objectives guided by small clean subsets may provide limited supervision, whereas leveraging the entire dataset for indirect supervision yields more stable training signals, particularly when clean annotations are scarce.

Importantly, MRMP achieves these results **without noise-specific tuning during final training**, including the use of clean validation sets, ensembling, or selective sample filtering. These findings highlight the robustness, generalizability, and practical efficiency of MRMP for downstream training under noisy supervision.

V. DISCUSSION

This section examines the behavior and effectiveness of the proposed MRMP framework by analyzing the accumulation of confirmation bias during the purification process, as well as the associated uncertainty and its potential utility for real-world human re-annotation. A detailed analysis of label correction dynamics is provided in the supplementary material (see Section S.5).

A. Analysis on Accumulated Confirmation Bias

Fig. 3 presents a comparative analysis of MRMP and representative noisy-label learning methods on CIFAR-10N

and CIFAR-100N, which contain human-annotated instance-dependent label noise. The analysis focuses on the accumulation of confirmation bias, quantified using two complementary metrics: 1) *Noise Detection Rate (NDR)*, defined as the proportion of detected noisy samples (i.e., samples whose labels are flipped during purification), and 2) *Correct Flip Rate (CFR)*, defined as the proportion of noise-to-clean corrections among all corrected samples. While NDR reflects the aggressiveness of noise detection, CFR directly characterizes the reliability of label correction and thus serves as an indicator of accumulated confirmation bias.

Across both datasets, MRMP consistently achieves high CFR while maintaining a moderate NDR, resulting in comparable or superior purified label accuracy (see Section III). This behavior indicates that MRMP effectively improves data quality while limiting the accumulation of confirmation bias. In contrast, several baselines, including DivideMix, ProMix, and WSC, exhibit strong noise detection capability but rely on aggressive correction strategies, as evidenced by very high NDRs accompanied by substantially reduced CFRs. In particular, on CIFAR-100N, most competing methods achieve CFR values below 50%, indicating that fewer than half of the applied label flips align with the underlying clean labels. Such low-precision corrections are prone to amplifying confirmation bias, posing a serious concern for practical data purification. By comparison, MRMP consistently maintains a high CFR, remaining **above 75%** even under severe noise, demonstrating reliable correction behavior in realistic noisy settings. Additional analysis of accumulated confirmation bias on CIFAR-10/100 is provided in the supplementary material (see Section S.6).

B. Semi-Automatic Correction on Remote Sensing Datasets

The high Correct Flip Rate (CFR) achieved by MRMP reflects the strong reliability of its purified datasets, which in turn enables practical downstream use cases—most notably, a second-stage human annotation process aimed at further improving label quality. To make this feasible, an efficient strategy for identifying potentially erroneous labels is required, underscoring the importance of uncertainty estimation. To explore this possibility, Fig. 4 examines the relationship between model confidence and prediction correctness, along with simulated correction curves for the UC Merced and AID datasets under 40% asymmetric noise. Despite moderate correlation levels (Pearson $r \approx 0.4$ for UC Merced and $r \approx 0.3$ for AID), the simulated correction process, which is performed by re-labeling samples in ascending order of confidence, reveals that correcting only 30% of the least confident predictions can correct the remaining noisy labels by around 85% on both datasets. These results support the feasibility of confidence-guided refinement, especially in domains where acquiring clean supervision is expensive. With its round-wise, soft-label trajectories and accumulated uncertainty signals, MRMP provides a robust foundation for enabling semi-automatic correction pipelines in expert-labeled datasets.

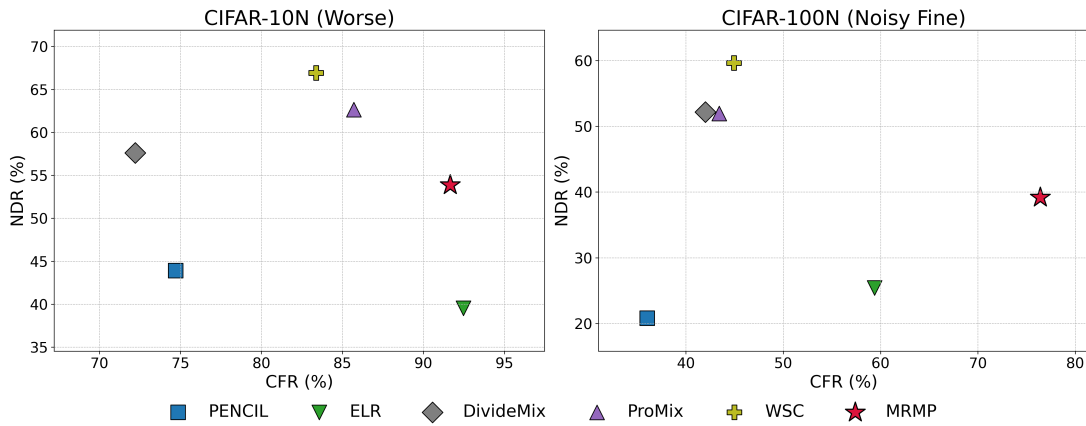


Fig. 3. Comparison of correct flip rate (CFR) and noise detection rate (NDR) for five noisy-label learning methods on CIFAR-10N (Worst) (top row) and CIFAR-100N (Noisy-Fine) (bottom row). Each point represents a method, with our proposed MRMP highlighted in pink using a star marker.

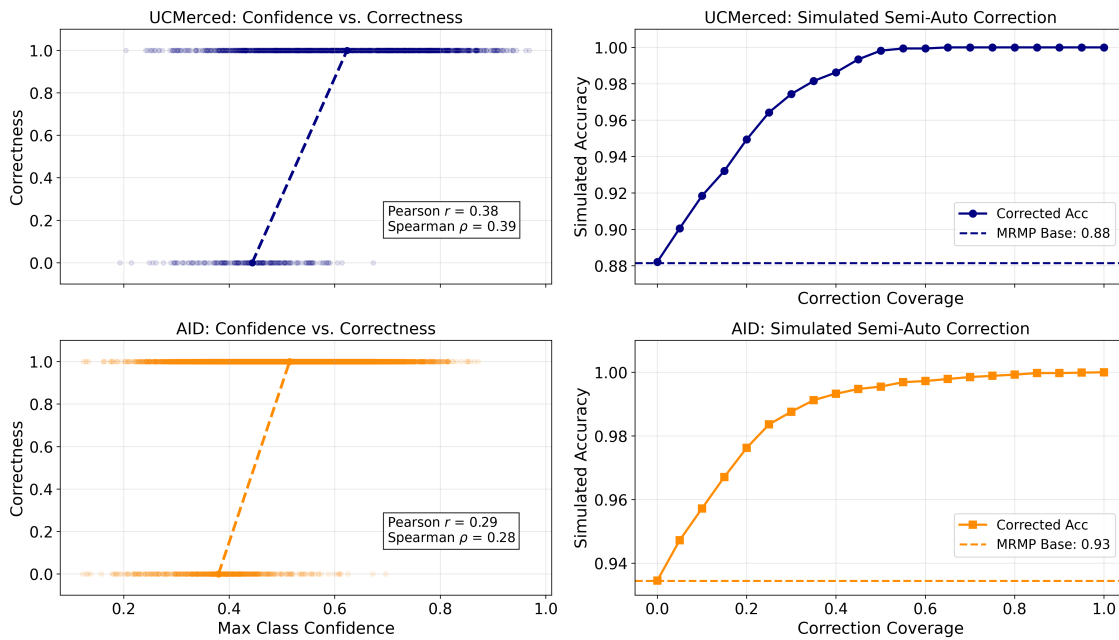


Fig. 4. Confidence-based uncertainty analysis and simulated semi-automatic annotation improvement on the UCMerced and AID datasets with 40% pairflip noise. (Left column) Relationship between maximum class confidence and prediction correctness. Dashed lines connect the average confidence of incorrect and correct predictions. Pearson correlation (r) and Spearman rank correlation (ρ) quantify the degree of alignment. (Right column) Simulated annotation accuracy assuming automatic correction of low-confidence samples. Dashed lines indicate baseline (uncorrected) model accuracy.

VI. ABLATION STUDY

To assess the contribution of each key component in the proposed MRML framework, we conduct ablation experiments on CIFAR-10 and UC Merced under 50% symmetric label noise, representing general-purpose and expert-annotated noise scenarios, respectively. Results are summarized in Table V.

Effect of Multi-round Purification: Removing the multi-round purification process and instead using normalized single-round predictions for label replacement results in substantial performance degradation (−7.82% on CIFAR-10, −15.18% on UC Merced). This confirms that iterative refinement is critical for progressively enhancing label quality and mitigating abrupt, unstable shifts in supervision.

Effect of KL-based Early Stopping: We disable the KL-

based stopping criterion and fix the purification rounds to 5. This change causes a moderate decline in accuracy (−2.34% on CIFAR-10, −0.77% on UC Merced), indicating that adaptive termination of purification helps avoid both under- and over-refinement of labels.

Effect of Exponential Moving Average (EMA): Excluding EMA results in a sharp accuracy drop on CIFAR-10 (−21.52%) and a moderate decline on UC Merced (−0.77%), underscoring the importance of temporal smoothing. Inspired by the success of ELR and ELR+ [18], EMA mitigates the gradual drift of model predictions from clean to noisy by stabilizing pseudo-labels across purification rounds—an effect that proves especially beneficial for maintaining robust label updates in MRMP on CIFAR-10.

TABLE V
ABLATION STUDY RESULTS (%) ON CIFAR-10 AND UC MERCED UNDER
50% SYMMETRIC LABEL NOISE.

Variant	CIFAR-10	UC Merced
MRML (full)	98.13	96.13
w/o Multi-round	90.31	80.95
w/o KL-based stopping	95.79	95.36
w/o EMA	76.61	95.36
w/o Meta-learning	90.09	83.93
w/o Pretrained model	74.05	62.38

Effect of Meta-Learning: Disabling meta-learning and performing label updates directly on the EMA outputs with a fixed momentum coefficient ($\beta = 0.2$) results in a performance drop of 8.04% on CIFAR-10 and 12.2% on UC Merced. This confirms that meta-optimization plays a crucial role in effectively distinguishing clean from noisy samples—even in the absence of clean validation data.

Effect of Pretraining: Replacing pretrained weights with random initialization degrades performance to 74.05% on CIFAR-10 and 62.38% on UC Merced. While the framework still outperforms the noisy-label baseline, this result underscores the importance of pretrained representations in guiding effective label purification. Without such prior knowledge, the model becomes more susceptible to the influence of noisy labels, leading to unstable and suboptimal refinement.

Overall, these results validate the necessity of each component in enhancing label reliability and highlight the central roles of multi-round refinement and meta-learning across both general-purpose and domain-specific (e.g., remote sensing) classification tasks.

VII. CONCLUSION

In summary, this work demonstrates the practical potential of clean-label-free purification through the proposed multi-round meta-purification (MRMP) framework. MRMP exhibits strong robustness to label noise across diverse benchmarks, including CIFAR-10/100, CIFAR-10N/100N, mini-Clothing-1M, and expert-annotated remote sensing datasets, showing that decoupling label refinement from model training is an effective strategy for improving label quality without clean supervision. Our theoretical analysis motivates a KL-based stopping criterion that identifies an appropriate refinement round and is empirically shown to reduce confirmation bias during label purification. In addition, the evolution of soft labels provides informative signals for identifying uncertain or error-prone samples, suggesting potential benefits for human-AI collaborative annotation.

Several challenges remain for practical implementation. Improving the scalability and adaptability of MRMP to diverse real-world datasets, including settings that require training from scratch or involve large-scale data and higher-capacity models, is an important direction for future work. Extending MRMP to multi-label problems and structured prediction tasks, such as semantic segmentation, also remains open. Overall, MRMP offers a promising step toward scalable and practical label purification in scenarios where high-quality annotations are costly or inconsistently available.

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VIII. BIOGRAPHY SECTION

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